

Hongyi Hao

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Education

Shanxi Institute of Technology
B.Eng. in Computer Science

Yangquan, China
Sep.2022 – Jul.2026

Columbia University
M.Sc. in Computer Engineering

New York City, USA
Aug.2026 – Present

Research Interests

Core Research Interests: Autonomous Agents, Embodied AI, Robot Learning, Reinforcement Learning, Artificial General Intelligence (AGI).

Core Research Question: How can we build more general intelligent systems that can understand, reason, plan, and make decisions autonomously in complex real-world environments, enabling agents and robots to reliably perform a wide range of tasks beyond narrowly defined domains?

Publications

Preprints

[P2] Cross-Embodied Reinforcement Learning Control Method for Quadruped Robots on Complex Terrain.

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ICRSS, 2025.

[P1] Structured Pruning for Efficient YOLO11 Deployment: A Comparative Study of LAMP and SPPF Algorithm.

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ICMAEE, 2025.

Journal Proceedings

[J1] EEGAuth: A Secure and Lightweight EEG-Based System Integrating Authentication and Key Generation.

Xun Han, Jun Xiao, Yifan Liu, Ruilin Zhang, Kaibiao Zhu, Hongyi Hao, Youqi Li, Fan Li, Qian Zhang.

IEEE Internet of Things Journal, 2025. [\[Paper\]](#)

Conference Proceedings

[C8] Research and Application of Multi-Agent Cooperative Decision Algorithm Based on Deep Reinforcement Learning.

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ACESPE, 2025. [\[Paper\]](#)

[C7] Multidimensional Data Query Processing Methods Using Artificial Intelligence and Machine Learning Algorithms.

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SSITCON, 2025. [\[Paper\]](#)

[C6] VoteRFID: Multi-Target Rotation Recognition Algorithm Based on Phase Encoding via Markov Transition Field.

Zhonghao Wang, Hongyi Hao, Biaokai Zhu, Yiran Li, Yaxuan Guo, Jiaqi He, Qian Zhang

IEEE ISPA, 2025. [\[Paper\]](#)

[C5] Temperature-Aware Liquid Sensing with mm-Wave Radar.

Zeyu Fan, Zhaokui Wang, Hongyi Hao, Zhao Lv, Ping Li

IEEE ISPA, 2025. [\[Paper\]](#)

[C4] Data Security Strategies and Technologies for Robust Cloud Computing.

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Discover Applied Science. [\[Paper\]](#)

[C3] Research on Multi-Robot Obstacle Avoidance Technology based on HRVO Combined with Deep Reinforcement Learning.

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ICCASIT, 2025. [\[Paper\]](#)

[C2] Stability Analysis of Polygonal Slopes Considering Spatial Variability of Soil Parameters.

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ICDA, 2025. [\[Paper\]](#)

[C1] ECHOES: Input Sensing and Reconstruction Model utilizing CNN and LSTM with Mobile Sensor Data.

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IEEE ICICC 2025. [\[Paper\]](#)

Research Experiences

Carnegie Mellon University

Pittsburgh, USA

Research Intern, Advised by Prof. [Soumya Kar](#)

May.2025 – Aug.2025

- **Research Topic:** We conducted online research seminars, systematically study distributed and federated learning, complete various machine learning tasks based on real data, propose a federated learning method with high communication efficiency, and explore its applications in autonomous driving and edge computing.

Internship Experiences

ArcherMind Technology

Nanjing, China

Data Analyst Intern

Jul.2023 – Aug.2023

- **Responsibility:** Collected, organized, and cleaned data generated by in-vehicle AI systems to remove unreasonable or noisy entries, ensuring accurate AI decision-making and responses while enhancing overall user satisfaction.

Institute of Automation, Chinese Academy of Sciences

Nanjing, China

Backend Development Intern

Jul.2024 – Sep.2024

- **Responsibility:** Performed basic backend development using Spring Boot, gained an understanding of MySQL and Oracle principles, explored Spring Cloud-based microservices architecture, and studied the workflow of distributed systems.

Projects

An Improved YOLO-Based Method for Tunnel Concrete Defect Detection & Robot

Shanxi, China

Founder & Developer. [\[Introduction\]](#)

Sep.2022 – Feb.2026

- A radar waveform recognition algorithm based on the YOLO11 framework was developed, marking the first large-scale application of machine learning algorithms in the field of civil engineering.
- Structural improvements were made to an adsorption-type tunnel inspection robot using the design concept of anti-gravity negative pressure robots.
- Have inspected **68 tunnels**, with a survey line length exceeding **190,000 meters**.

AI-based Expressway Strip Litter Collection Robot

Shanxi, China

Developer. [\[Code\]](#)

Jul.2025 – Feb.2026

- High-precision identification of waste in complex environments based on an improved YOLOv11 model.
- Enabled continuous model improvement by incorporating newly collected data during deployment for iterative retraining and accuracy enhancement.

Acoustic-based gaze recognition framework

Shanxi, China

Developer. [\[Code\]](#) [\[Reference\]](#)

Mar.2025 – Jul.2026

- Reproduced the core pipeline of the [GazeTrak system](#) based on acoustic sensing for eye-tracking, including signal processing and deep learning-based gaze estimation.
- Improvements to model training and data processing were proposed and evaluated, achieving greater robustness and accuracy under various complex conditions.

Award & Honors

2025 **National 2nd Prize, College Students' Extracurricular Academic and Scientific Works Competition.**
2025 **Provincial 2nd Prize, Shanxi Province Small Innovation Competition.**
2025 **Silver Award, China International College Student Innovation Competition.**
2025 **Provincial Special Award, College Students' Extracurricular Academic and Scientific Works Competition.**
2024 **Provincial Winner, Shanxi Small and Medium Enterprises Innovation and Entrepreneurship Competition.**
2023 **Provincial 2nd Place, Shanxi Central Youth City Innovation and Entrepreneurship Competition.**
2023 **Provincial 3rd Prize, Zhongkong Xinda Cup Computer Programming Competition for College Students.**
2026 **Awarded the "Outstanding Graduate" Title by the College.**
2025 **Awarded the "Outstanding Student" Title by the College.**
2025 **Department 2nd Class Scholarship, Academic Year 2024-2025.**
2024 **College Individual Scholarship, Academic Year 2023-2024.**
2024 **Department 3rd Class Scholarship, Academic Year 2023-2024.**
2023 **Department 2nd Class Scholarship, Academic Year 2022-2023.**

Skills

Programming: C/C++, Java, Python, SQL, JavaScript, TypeScript, NumPy, Linux, Git, Android, PHP, MATLAB

Framework: OpenCV, ROS, Pytorch, Spring Boot, YOLO, Flask, Qt, VUE, Docker, Cmake, Node.js

Hardware: Raspberry Pie, Walnut Pie, Arduino